CCIThink

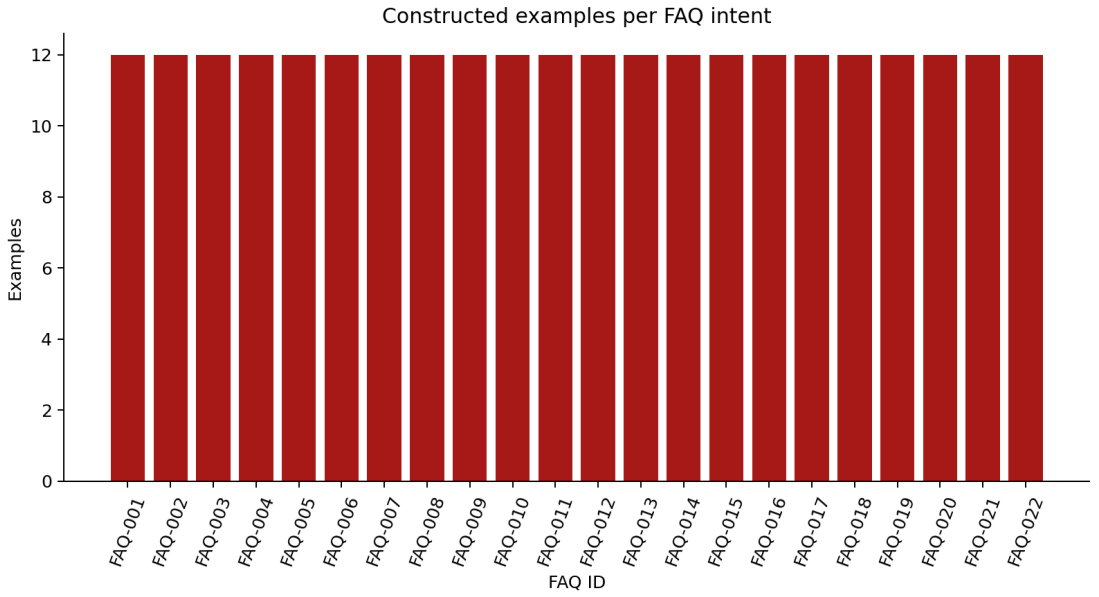
- CCIT information chatbot + explainable FAQ classification
- Boncajes, Rhian Jared D.
- Marcos, Maricar N.
- Carandang, Jamie Pearl U.
- Principles of Artificial Intelligence and Data Science · November 2026

Problem and objectives

- Find answers within 22 documented department FAQs.
- Preserve the existing chat UI and source-support checks.
- Demonstrate data preparation, learning, evaluation and interpretation.

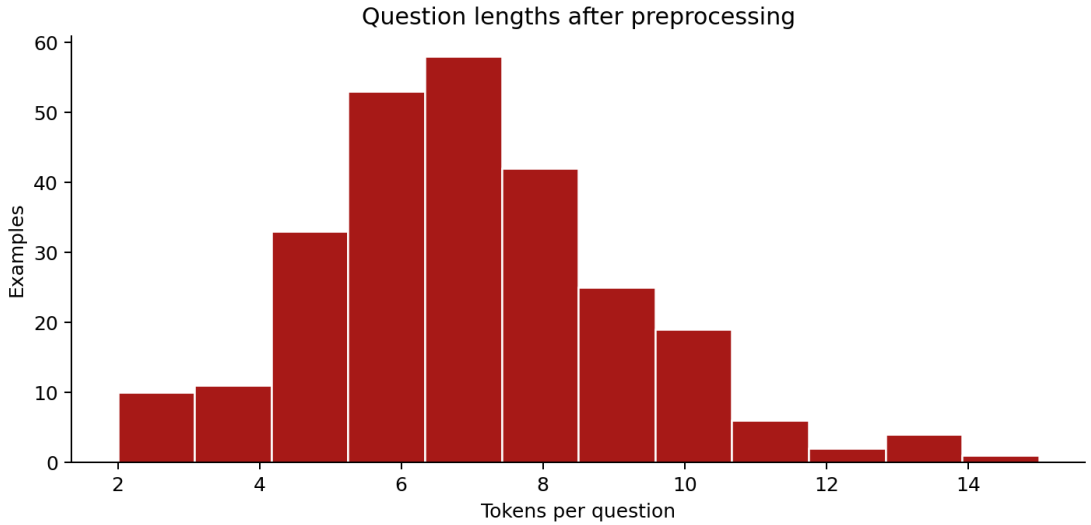
Dataset and provenance

- 264 AI-assisted constructed questions; 22 FAQ target labels.
- 176 training / 44 validation / 44 test; fixed assignments.
- Not real student conversations; no private records.
- Eight training, two validation and two test questions per label.



Preprocessing and EDA

- Validate schema, label IDs, split membership and input limits.
- Reject label conflicts and cross-split duplicates.
- 264 normalized; 0 missing, duplicate or invalid rows removed.
- Mean length 7.08 tokens; 7 longer valid questions retained.

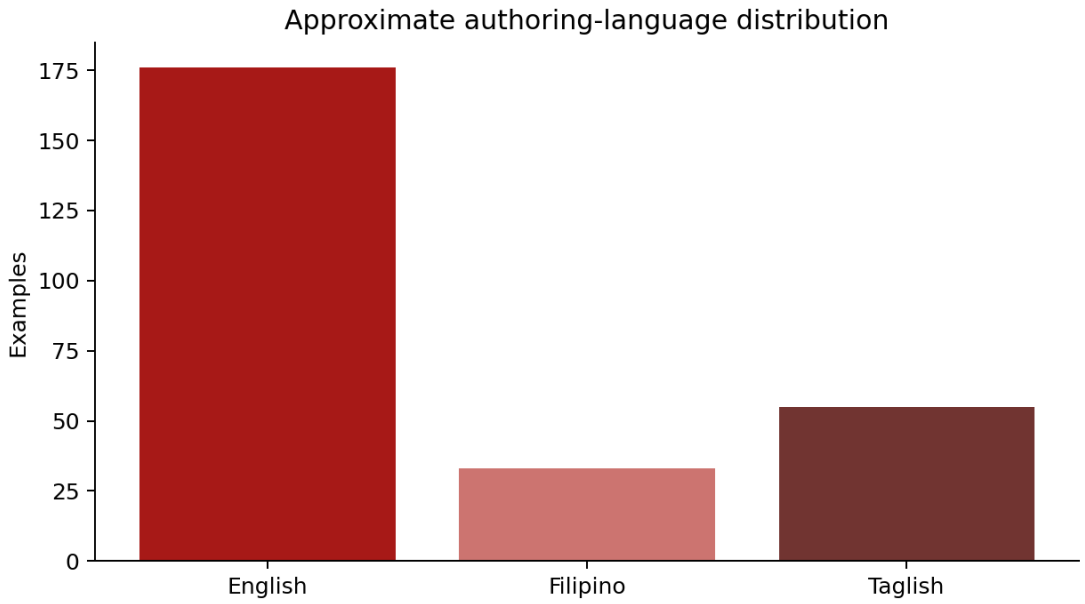


Language coverage

• 176 English / 33 Filipino / 55 Taglish authoring tags.

• Tags are approximate, not independently annotated.

• The corpus is balanced by FAQ, not sampled from student demand.

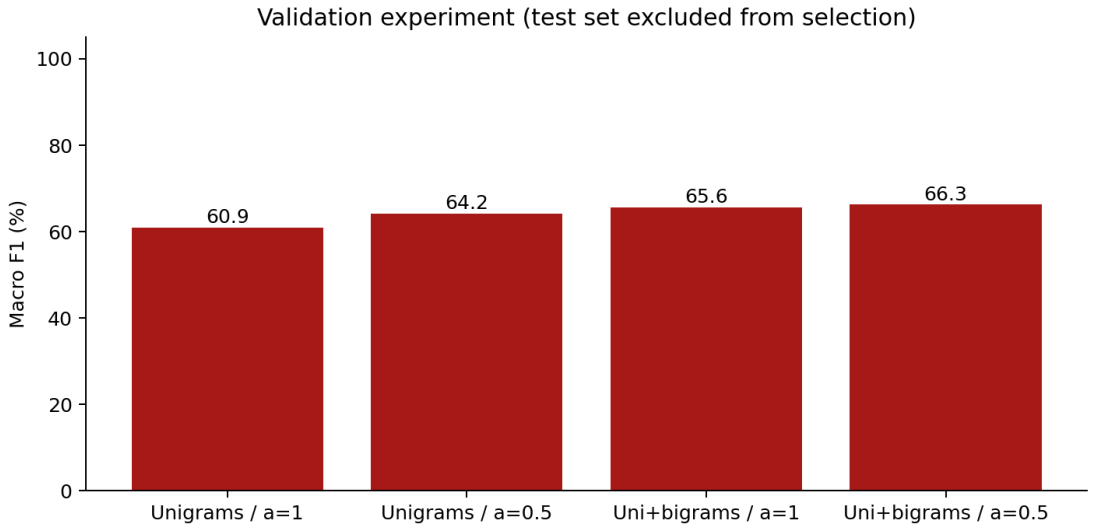


Implemented algorithm

- Multinomial Naive Bayes, implemented in standard Python.
- Count unigrams or unigrams plus adjacent bigrams.
- Learn vocabulary and class-feature counts on training data only.
- Additive smoothing avoids zero likelihoods; scores are uncalibrated.

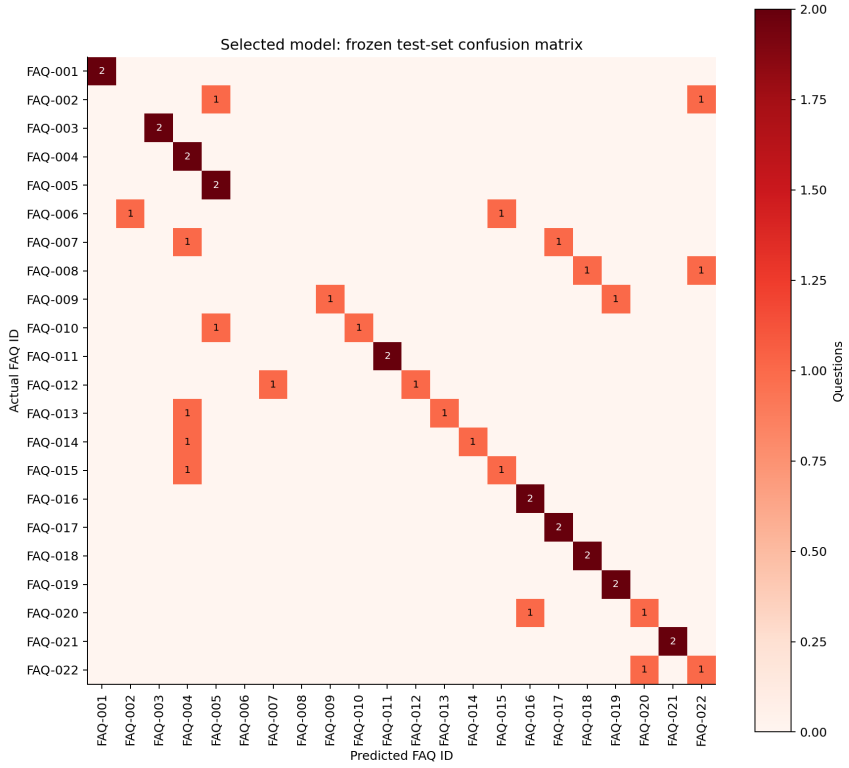
Validation experiment

- Four fixed feature/smoothing configurations.
- Choose using validation macro F1, not test accuracy.
- Selected: unigrams + bigrams, alpha = 0.5.
- Validation: 31/44 correct; macro F1 = 0.663.



Frozen-test results

- 28/44 correct: 63.6% accuracy.
- Macro F1 = 0.595; macro precision = 0.629.
- Majority-class baseline: 2/44 correct (4.5%).
- 16 mistakes; only two test questions per label.



Error analysis

- English-tagged test: 16/22 correct.
- Filipino-tagged test: 3/8; Taglish: 9/14.
- Overview, offerings and IT/CS definitions were difficult.
- Small constructed groups do not establish population differences.

Working demonstration

- Run python analyze.py, then python app.py.
- Use /analysis for charts, cleaning, metrics and predictions.
- Use / for the preserved chat interface and short bold answers.
- ML only adds a bounded ranking bonus after source checks.
- 263 automated tests; 26 browser scenarios; 95/95 chat checks.

Ethics and limitations

- No private student records; no stored chat transcript.
- Department facts and source permissions still need appropriate review.
- AI assistance is disclosed; every author must understand the code.
- No participant study, production-load result or successful live LLM claim.

Conclusion and next steps

- A reproducible AI/Data Science workflow now supports the chatbot.
- The classifier outperformed a weak baseline but needs improvement.
- Next: confirmed facts, permissioned student examples, independent labels and a fresh test set.
- Questions and defense discussion